

Morphisms — Prime Structure Math-Clean Specification

Version: 1.0

Status: Normative

0. Scope

This specification defines the morphisms that connect the boundary, schedule, response, and signature layers of the Prime Structure. It is mathematically self-contained and testable. Core invariants: **48/256/12 288, R96, 3/8**.

1. Core sets and actions

1.1 Index, selectors, channels

- Index cycle: $I = \mathbb{Z}/768$.
- Selectors: $S = \mathbb{Z}/256$.
- Channels: $C = \{0, 1, 2\}$ with orientation carried by C .

A **fair schedule** is a bijection $\sigma : I \rightarrow S \times C$, written $\sigma(i) = (s(i), c(i))$, such that each $s \in S$ appears exactly three times and each channel appears exactly 256 times.

1.2 Boundary subgroup

Let

$$G^\circ = U(48) \times U(256), \quad |U(48)| = 16, |U(256)| = 128, |G^\circ| = 2048.$$

Concrete generators: - $U(48)$ generated by $\{-1, 5, 17\}$ modulo 48.

- $U(256)$ generated by $\{-1, 5\}$ modulo 256.

Action. G° acts on $S \times C$ by $(a, b) \cdot (s, c) = (a s \bmod 256, c)$. The action preserves channel orientation.

1.3 Response parameters and the R96 map

Parameters: positive reals $\alpha_0, \dots, \alpha_8$ with $\alpha_0 = 1$ and $\alpha_4 \alpha_5 = 1$. Define $T = \{1, 2, 3, 6, 7, 8\}$. For $b \in \{0, 1\}^T$ and $d \in \{-1, 0, 1\}$,

$$R(b, d) = \left(\frac{\alpha_4}{\alpha_5}\right)^d \prod_{i \in T} \alpha_i^{b_i}.$$

Genericity. Assume $\{\log(\alpha_4/\alpha_5)\} \cup \{\log \alpha_i : i \in T\}$ is $\mathbb{Q}$ -independent. Then $|\text{Im } R| = 3 \cdot 2^5 = 96$ (R96), giving **3/8** compression from 256 selectors to 96 classes.

Define the **response semiring** $C_{96} \cong \mathbb{Z}_3 \times B^5$ (componentwise operations; $B = \{0, 1\}$ with XOR/AND). The crush map $C_{96} \rightarrow B$ sends $(d, \mathbf{b}) \mapsto (d \text{ / } 0) \nabla (\mathbf{b} \text{ / } \mathbf{0}) :=$

2. Protocol morphisms

We define three morphisms between layers:

1. **Compression** $\kappa : I \rightarrow C_{96}$.
2. **Klein window** $\omega : C_{96} \rightarrow \mathcal{W}$ to window classes.
3. **Signature** $\chi : C_{96} \rightarrow \{0, 1\}^{10}$ with canonical lifts to $\{0, 1\}^{12}$.

2.1 Compression morphism κ

Let $\Pi_S : S \rightarrow \{0, 1\}^T$ assign a toggle vector $b(s)$. Let $\Pi_C : C \rightarrow \{-1, 0, 1\}$ assign the pair variable $d(c)$. Define

$$\kappa(i) = \iota(d(c(i)), \bar{b}(s(i))) \in C_{96},$$

where $\bar{b}$ denotes the 5-dimensional parity class of b and $\iota : \mathbb{Z}_3 \times \mathbb{Z}_2^5 \hookrightarrow C_{96}$ is the canonical embedding. Under genericity, distinct pairs $(d, \bar{b})$ yield distinct responses.

Equivariance. For $(a, b) \in G^\circ$ and any rotation $h_r : i \mapsto i + r$ of I , there exist automorphisms $\theta_a \in \text{Aut}(C_{96})$ and $\rho_r \in \text{Aut}(C_{96})$ such that

$$\kappa((a, b) \cdot i) = \theta_a \kappa(i), \quad \kappa(h_r i) = \rho_r \kappa(i).$$

2.2 Klein windows and ω

Let $V_4 = \{e, x, y, xy\}$ act on C_{96} by flipping two independent parity coordinates in B^5 (componentwise XOR on the chosen coordinates; $\mathbb{Z}_3$ factor fixed). For $x \in C_{96}$, its **Klein window** is the orbit

$$\mathcal{W}(x) = \{v \cdot x : v \in V_4\}, \quad 1 \leq |\mathcal{W}(x)| \leq 4.$$

Define $\omega : C_{96} \rightarrow \mathcal{W}$ by $\omega(x) = \mathcal{W}(x)$. This is a homomorphism of V_4 -sets and is constant on V_4 -orbits.

Compatibility. For any $i \in I$, $\omega \circ \kappa(i)$ is invariant under flips of the designated parity pair. Hence windows are stable summaries for parity-pair symmetries.

2.3 Signature morphism χ and NF-Lift

Let $S \in \mathbb{F}_2^{12 \times 10}$ be the fixed full-rank toggle-signature matrix

$$S = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Define $\Phi(x) = S^\top x \in \{0, 1\}^{10}$ for $x \in \{0, 1\}^{12}$. Fibers are affine cosets of $\ker S$ of size $2^{12-10} = 4$.

Define $\chi : C_{96} \rightarrow \{0, 1\}^{10}$ by composing a fixed encoding $\varepsilon : C_{96} \rightarrow \{0, 1\}^{12}$ with $\Phi : \chi = \Phi \circ \varepsilon$. The encoding ε selects twelve binary features of C_{96} (five parity bits, one tri-state bit expanded to two binaries, and five auxiliary toggles) in a fixed order.

NF-Lift. Define $\text{lift} : \{0, 1\}^{10} \rightarrow \{0, 1\}^{12}$ as the lexicographically smallest solution of $Sx = y$ using a fixed RREF pivot order. Round-trip holds: $\Phi(\text{lift}(y)) = y$. Each fiber is a torsor under $\ker S$.

3. Holonomy and commutation laws

3.1 Schedule holonomy

For $r \in \mathbb{Z}/768$, define $h_r : i \mapsto i + r$. Then

$$\sigma \circ h_r = (h'_r \times h''_r) \circ \sigma,$$

where $h'_r(s) = s + pr \bmod 256$ and $h''_r(c) = c + qr \bmod 3$ for strides $p \in U(256)$, $q \in U(3)$ used by the constructor in §5. Consequently, $\kappa \circ h_r = \rho_r \circ \kappa$ for an automorphism ρ_r induced by (p, q) .

3.2 Commutative diagram

For all $i \in I$:

$$I \xrightarrow{\kappa} C_{96} \downarrow h_r \quad \downarrow \rho_r I \xrightarrow{\kappa} C_{96} \quad C_{96} \xrightarrow{\omega} \mathcal{W} \downarrow \theta_a \quad \downarrow \bar{\theta}_a C_{96} \xrightarrow{\omega} \mathcal{W}$$

Here $\bar{\theta}_a$ is the induced action on windows. Similarly $\chi \circ \theta_a = \chi$ when θ_a flips only coordinates annihilated by S .

4. Basis size 12 288

Let a **basis** $\text{Idx} \subseteq I \times S \times C$ generate all schedules under G° while preserving orientation and fairness. Then $|\text{Idx}|$ is a multiple of $3 \cdot 256$ and of 48. The lower bound is $\text{lcm}(3 \cdot 256, 48) = 12\,288$. A representative-per-orbit construction attains this, hence the bound is tight.

5. Deterministic constructors and artifacts

- **Schedule constructor:** $\sigma(t) = (5t \bmod 256, 2t \bmod 3)$.
- **Boundary generators:** $U(48) = \langle -1, 5, 17 \rangle$, $U(256) = \langle -1, 5 \rangle$.
- **Encoding ε :** fixed bit order and feature map documented alongside test artifacts.
- **NF-Lift:** fixed pivot order $(1, 2, \dots, 10)$, lexicographic tie-break.

6. Conformance tests

6.1 Boundary

- Enumerate orders of generators and verify $|U(48)| = 16$, $|U(256)| = 128$, product 2048.
- Verify equivariance of κ under $a \in U(256)$.

6.2 Schedule

- Check bijection and fairness counts.
- Verify holonomy identities with strides $(p, q) = (5, 2)$.
- Confirm rotation-invariant means and variances by closed-form sums.

6.3 R96 and compression

- Evaluate $R(b, d)$ for all $d \in \{-1, 0, 1\}$, $b \in \{0, 1\}^T$; assert cardinality 96.
- Confirm pair normalization invariance under $(\alpha_4, \alpha_5) \mapsto (t\alpha_4, t^{-1}\alpha_5)$.

6.4 Klein windows

- Validate V_4 -action on C_{96} and that ω is constant on orbits.
- Check window sizes ≤ 4 across the corpus.

6.5 Signatures and lifts

- Compute $\text{rank } S = 10$ and $|\ker S| = 4$.
- Verify $\Phi(\text{lift}(y)) = y$ for all y .
- Confirm that χ is stable under flips in coordinates annihilated by S .

7. Notes on implementation

- Use bit-packed linear algebra over $\mathbb{F}_2$ for S and NF-Lift.
- Implement group actions via modular multiplication with generator exponentiation tables.
- Enforce the $\mathbb{Q}$ -independence genericity for α by rank tests on sampled integer relations.

This document defines κ , ω , and χ , proves their compatibilities with boundary and schedule actions, and provides complete acceptance criteria for the morphism layer of the Prime Structure.